Business Requirements Document (BRD)

E-Commerce Enterprise Data Warehouse & Analytics Platform

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1. Introduction

This Business Requirements Document (BRD) outlines the operational and analytical mandates required to establish an enterprise-grade E-Commerce Data Warehouse and Business Intelligence platform. The project aims to consolidate disparate operational transaction records into a centralized analytics solution to support executive decision-making, revenue growth, and operational optimization.

2. Business Objectives

The primary strategic goals of this initiative are centered around establishing commercial clarity, pricing integrity, and customer tier understanding:

Strategic Goal	Business Purpose	Target Value
Commercial Visibility	Centralize all transactional revenue, order volumes, and average basket sizes into a single source of truth.	100% Sales Reconciliation
Pricing Integrity	Monitor pricing variance between actual checkout prices paid and base standard catalog list prices.	Margin Erosion < 2%
Customer Segmentation	Analyze lifetime purchasing behavior and order frequency split by loyalty tiers (Silver vs. Platinum).	Platinum Retention (+10%)
Order Tracking	Evaluate order lifecycle fulfillment statuses (`placed`, `shipped`, `delivered`, `cancelled`).	Cancellation Rate (<3%)

3. AS-IS Assessment (Current State)

Currently, operational transactional data is isolated across four separate, static flat files (`users.csv`, `products.csv`, `orders.csv`, and `order_items.csv`). The organization suffers from manual spreadsheet manipulation, lack of automated cross-file joining, absence of dynamic filtering across time/categories, and delayed reporting turnaround times.

4. TO-BE Vision (Future State)

The target future state introduces an automated, scalable cloud data platform built inside Microsoft Fabric. Raw operational extracts will be continuously ingested into a star schema data warehouse, surfacing real-time KPI metrics and interactive visual analytics to business stakeholders via a 3-sheet executive dashboard.

5. Business Requirements

The core business requirements define the operational rules and analytical capabilities expected from the system:

Req ID	Requirement Name	Detailed Business Description
BR-01	Automated Data Ingestion	System must automatically ingest source CSV extracts without requiring manual spreadsheet formulas.
BR-02	Historical Order Logic	System must enforce strict business logic ensuring order timestamps occur strictly after user join dates.
BR-03	Price Variance Calculation	System must track variance between actual checkout prices (`unit_price`) and base catalog prices (`base_price`).
BR-04	Multi-Dimensional Slicing	Business users must be able to dynamically filter analytics by Date Range, Order Status, Category, and Tier.

6. Data and Reporting Requirements

The reporting environment must feature an interactive 3-sheet Business Intelligence dashboard providing tailored perspectives:

Dashboard Sheet	Key Performance Indicators (KPIs)	Core Analytical Charts
1. Sales & Revenue	Total Net Revenue, Total Orders Placed, Average Order Value (AOV).	Monthly Revenue/Order Trend, Revenue by Order Status, Daily Revenue by Day of Week.
2. Product Analytics	Total Units Sold, Average Selling Price (ASP), Total Price Variance.	Top 10 Products by Revenue, Category Revenue vs. Units, Base Price vs. Actual Price Variance.
3. User Insights	Active Purchasing Customers, Revenue per Customer, Order Frequency.	Revenue Share by Loyalty Tier, Customer Join Cohort Spending, Basket Size by Loyalty Tier.

> **Interactive Reporting Mandate**
> All dashboard sheets must support interactive cross-highlighting, allowing users to select specific categories or tiers and immediately update corresponding KPI cards and trend lines.

7. Scope and Constraints

Clear boundaries and operational constraints are established to ensure structured delivery:

Domain	Scope Definition & Technical Constraints
In-Scope	Ingestion of `users`, `products`, `orders`, `order_items` CSVs; Star Schema modeling; DAX calculations; 3-Sheet Power BI report.
Out-of-Scope	Real-time streaming ingestion, third-party ERP integrations, predictive machine learning models, mobile native app design.
Constraints	All historical order timestamps must strictly follow user creation dates; solution must utilize Microsoft Fabric ecosystem.

8. Success Criteria

Project sign-off and business acceptance will be validated against explicit quantitative criteria:

Criterion	Acceptance Benchmark	Validation Method
Data Accuracy	100% financial metric alignment between raw source CSV files and reporting measures.	Automated Audit Scripts
User Acceptance	Formal sign-off from Executive Leadership and Merchandising teams during UAT testing.	UAT Test Scripts
Performance	Dashboard visualizations render and update within < 2.5 seconds upon filter selection.	Performance Analyzer
Deployment	Successful publish of Star Schema Warehouse and Semantic Model to Fabric workspace.	Deployment Checklist

9. Approval and Sign-Off

By signing below, the key executive stakeholders approve the business requirements and scope detailed in this document.

VP of Executive Analytics

Name: Marcus Vance | Date: _____

Head of E-Commerce Operations

Name: Sarah Jenkins | Date: _____

Lead Enterprise Architect

Name: David Chen | Date: _____